

Quantum Mechanics

Q1. A bullet of mass 40gms and an electron both travel at the velocity of 1100m/sec. What wavelengths can associate with them? Why the wave nature of bullet is not revealed through diffraction effect?

Given:- Mass of an electron = 9.1×10^{-31} kg

Mass of bullet = 40×10^{-3} Velocity of bullet = 1100m/sec

Formula: - $\lambda = h/mv$

Solution: - Let wavelength associated with bullet be λ_1

$$\lambda_1 = h/mv$$

$$= \frac{6.63 \times 10^{-34}}{40 \times 10^{-3} \times 1100}$$
$$= 1.5068 \times 10^{-35} \text{ m}$$

Let wavelength associated with electron be λ_2

$$\lambda_2 = h/mv$$

$$= \left(\frac{6.63 \times 10^{-34}}{9.1 \times 10^{-31} \times 1100} \right)$$
$$= 6.623 \times 10^{-7} \text{ m}$$

$$\lambda_2 = 6623 \text{ \AA}$$

Q2. Electrons accelerated through 100V are reflected from crystal. What is the glancing angle at which the first reflection occurs? Lattice spacing = 2.15 Å

Given: - $V = 100 \text{ V}$

Formula: - $\lambda = \frac{12.26}{\sqrt{V}}$

Solution: - Wavelength associated with electrons is

$$\lambda = \frac{12.26}{\sqrt{V}}$$

$$\lambda = \frac{12.26}{\sqrt{100}}$$

$$= 1.226 \text{ \AA}$$

Now using Bragg's law for first order:

$$\lambda = 2d \sin \theta$$

$$1.226 \text{ \AA} = 2 \times 2.15 \times \sin \theta$$

$$\theta = 16.5658$$

Q3. Calculate the kinetic energy of an electron whose de broglie wavelength is 5000 \AA.

Given:- Mass of electron = $9.1 \times 10^{-31} \text{ kg}$

$$h = 6.625 \times 10^{-34} \text{ J.sec}$$

Formula:- $\lambda = h / \sqrt{2mE}$

Solution:- $E = \frac{h^2}{2m\lambda^2}$

$$= \left(\frac{(6.625 \times 10^{-34})^2}{2 \times 9.1 \times 10^{-31} \times (5000 \times 10^{-10})^2} \right)$$

$$= 9.6463 \times 10^{-25} \text{ J} = 6.021 \times 10^{-6} \text{ eV}$$

Q4. Calculate the velocity and the de broglie wavelength of α particles of energy 1 keV.

Given:- Mass of α particle = 6.68×10^{-27}

Formula:- $\lambda = h / \sqrt{2mE}$

Solution:-

$$E = 1\text{keV} = 1 \times 10^3 \times 1.6 \times 10^{-19} \text{ J}$$

$$\lambda = h/\sqrt{2mE}$$

$$= \frac{6.63 \times 10^{-34}}{\sqrt{2} \times 6.68 \times 10^{-27} \times 1 \times 10^3 \times 1.6 \times 10^{-19}}$$

$$= 0.4386 \times 10^{-9} \text{ m}$$

Now,

$$\lambda = h/mv$$

$$V = \frac{6.63 \times 10^{-34}}{6.68 \times 10^{-27} \times 4.534 \times 10^{-13}}$$

$$V = 0.2189 \times 10^6 \text{ m/sec}$$

Q5. What is the wavelength of an electron moving at $5.31 \times 10^6 \text{ m/sec}$?

Given: mass of electron = $9.11 \times 10^{-31} \text{ kg}$

$$h = 6.626 \times 10^{-34} \text{ J}\cdot\text{s}$$

Formula: - $\lambda = h/mv$

Solution: -De Broglie's equation is

$$\lambda = h/mv$$

$$= \frac{6.626 \times (10)^{-34}}{9.11 \times (10)^{-31} \times 5.31 \times (10)^6}$$

$$\lambda = 1.37 \times 10^{-10} \text{ m}$$

$$\lambda = 1.37 \text{ \AA}$$

Q6. A fast moving neutron has de-broglie wavelength 2×10^{-12} m associated with it. Find the following:

(a) Kinetic energy

(b) Phase and group velocity

($m_{\text{neutron}} = m_n = 1.675 \times 10^{-27}$ kg)

Solution: -

$$p = \frac{h}{\lambda} = \frac{6.63 \times 10^{-34}}{2 \times 10^{-12}} = 3.32 \times 10^{-22}$$

$$\therefore KE = p^2 / 2m = \frac{(3.32 \times 10^{-22})^2}{2 \times 1.675 \times 10^{-27}} = 3.280 \times 10^{-17}$$

since the velocity of the particle is same as the group velocity, the group velocity can be given as,

$$v_{\text{group}} = p/m = \frac{3.32 \times 10^{-22}}{1.675 \times 10^{-27}} = 1.98 \times 10^5$$

now using the relation

$$(v_{\text{group}})(v_{\text{phase}}) = c^2$$

$$\therefore v_{\text{phase}} = \frac{c^2}{v_{\text{group}}} = \frac{(3 \times 10^8)^2}{1.98 \times 10^5} = 4.55 \times 10^{11} \text{ m/sec}$$

Q7. A particle of mass 1.57×10^{-30} kg and kinetic energy 80 eV. Find the de Broglie wavelength, group and phase velocity of that wave.

Solutions: -

$$\lambda = \frac{h}{\sqrt{2mE}} = \frac{6.63 \times 10^{-34}}{\sqrt{2 \times 1.57 \times 10^{-30} \times 80 \times 1.6 \times 10^{-19}}} = 1.128 \times 10^{-10} \text{ m}$$

Now,

$$V_{\text{group}} = \frac{p}{m} = \frac{h}{\lambda m} = \frac{6.63 \times 10^{-34}}{1.218 \times 10^{-10} \times 1.57 \times 10^{-30}}$$

$$V_{\text{group}} = 4.7 \times 10^6 \text{ m/sec}$$

$$V_{\text{phase}} = \frac{c^2}{v_{\text{group}}} = \frac{(3 \times 10^8)^2}{4.7 \times 10^6} = 1.9149 \times 10^{10} \text{ m/sec}$$

Q8. The speed of electron is measured to within an uncertainty of $2 \times 10^{-8} \text{ m/s}$. what is the minimum space required by electron to be confined in an atom?

Solution: -

Using eq of uncertainty

$$\Delta x \cdot \Delta p_x \geq \frac{h}{2\pi}$$

$$\begin{aligned} \therefore \Delta x &\geq \frac{h}{2\pi \Delta p_x} \\ &\geq \frac{h}{2\pi \Delta v_x} \geq \frac{6.63 \times 10^{-34}}{2\pi \times 9.1 \times 10^{-31} \times 2 \times 10^4} \\ &\geq 5.7978 \times 10^{-9} \text{ m} \end{aligned}$$

Q9. Find the values of momentum and energy of a neutron in one dimensional box with impenetrable wall.

Given: - $m_n = 1.67 \times 10^{-27} \text{ kg}$, $L = 1 \text{ A}$

Formula:- $E_n = \frac{p_n^2}{2m}$, $E_n = \frac{n^2 h^2}{8mL^2}$

$$n = 1, 2, 3, \dots$$

Solution: -

$$\begin{aligned} E_n &= \frac{n^2 h^2}{8mL^2} \\ &= \frac{n^2 (6.63 \times 10^{-34})^2}{8 \times 1.67 \times 10^{-27} \times (10^{-10})^2} \\ &= \frac{n^2 (6.63 \times 10^{-34})^2}{(8 \times 1.67 \times 10^{-27} \times 10^{-10})^2} \end{aligned}$$

$$E_n = n^2 \times (3.32 \times 10^{-21}) \text{ J}$$

$$E_1 = (3.32 \times 10^{-21}) \text{ J}$$

$$E_2 = (13.28 \times 10^{-21}) \text{ J}$$

$$E_3 = (29.88 \times 10^{-21}) \text{ J and so on}$$

$$\frac{p_n^2}{2m} = \frac{n^2 h^2}{8mL^2}$$

$$p_n = \frac{nh}{2L} = \frac{n \times 6.63 \times 10^{-34}}{2 \times 10^{-10}}$$

$$= n (3.315 \times 10^{-24}) \text{ kg m/sec}$$

$$p_1 = 3.315 \times 10^{-24} \text{ kg m/sec}$$

$$p_2 = 6.63 \times 10^{-24} \text{ kg m/sec}$$

$$p_3 = 9.945 \times 10^{-24} \text{ kg m/sec and so on}$$

Q10. An electron moves in the x-direction with a speed of $1.88 \times 10^8 \text{ m/s}$. If this speed is measured to a precision of 1% with what precision can you simultaneously measure its position?

Solution: -

Uncertainty in measurement of speed

$$\Delta v = \frac{1.88 \times 10^8}{100} = 1.88 \times 10^6 \text{ m/s}$$

According to Heisenberg's uncertainty principle

$$\Delta x \cdot \Delta p \geq \hbar$$

$$\geq \frac{h}{2\pi}$$

$$\Delta x = \frac{h}{2\pi \Delta p}$$

$$\Delta x = \frac{h}{2\pi m \Delta v}$$

$$\Delta x = \frac{6.63 \times 10^{-34}}{2\pi \times 9.1 \times 10^{-31} \times 1.88 \times 10^6}$$

$$\Delta x = 6.1679 \times 10^{-9} \text{ m} = 6.1679 \text{ nm}$$

Q11. A hydrogen atom has diameter of $0.53 \text{ \AA}$. Estimate the minimum energy an electron can have in this atom.

- (a) Calculation of minimum momentum of an electron
According to Heisenberg's uncertainty principle

$$\Delta x \cdot \Delta p \geq \hbar$$

$$\Delta x_{\max} \cdot \Delta p_{\min} = \frac{\hbar}{2\pi}$$

$$\begin{aligned} \Delta p_{\min} &= \frac{\hbar}{2\pi \Delta x_{\max}} \\ &= \frac{6.63 \times 10^{-34}}{2\pi \times 0.53 \times 10^{-10}} \end{aligned}$$

$$\Delta p_{\min} = 1.9909 \times 10^{-24} \text{ kgm/s}$$

- (b) Calculation of minimum of energy of electron

$$\text{Kinetic energy (E)} = \frac{1}{2}mv^2 = \frac{p^2}{2m}$$

$$\begin{aligned} \text{Kinetic energy (E}_{\min}) &= \frac{p_{\min}^2}{2m} \\ &= \frac{(9.94 \times 10^{-25})^2}{2 \times 9.1 \times 10^{-31}} \\ &= 5.429 \times 10^{-19} \text{ J} \end{aligned}$$